Agilex 5 Camera 4Kp60 HDR

Lab Bring-up & Day-to-Day Run Guide

Design: 4Kp60 Multi-Sensor HDR Camera Solution (rel/26.1)

Board: Agilex 5 FPGA E-Series 065B Modular Development Kit

Prepared for: Sahil / lab use

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NOTE: This guide captures the working procedure from your bring-up session, including MTU workaround and serial/GUI details that are easy to forget.

1. Design Links

- Overview: https://altera-fpga.github.io/rel-26.1/embedded-designs/agilex-5/e-series/modular/camera/camera_4k/camera_4k/
- Detailed doc: https://github.com/altera-fpga/agilex5-ed-camera/blob/rel/26.1/docs/es/camera/camera_4k/camera_4k.md
- Repo: https://github.com/altera-fpga/agilex5-ed-camera (branch rel/26.1)
- Release binaries: https://github.com/altera-fpga/agilex5-ed-camera/releases/tag/rel-26.1-isp_hdr-MDK_RevB_GrpB
- Quick start / build: https://github.com/altera-fpga/agilex5-ed-camera/blob/rel/26.1/AGX_5E_Altera_Modular_Dk_ISP_designs/HDR_CAMERA.md

2. One-time Programming (already done)

You only need this again if you wipe the microSD or QSPI flash.

Files used

- top.core.jic - QSPI boot image
- hps-first-vvp-isp-demo-image-agilex5_mk_a5e065bb32aes1.wic(.gz) - microSD image

microSD

```
gunzip -k hps-first-...wic.gz
sudo umount /dev/sdX1   # if mounted
sudo dd if=hps-first-....wic of=/dev/sdX bs=1M status=progress conv=fsync
sync && sudo eject /dev/sdX
```

QSPI (once)

- Power off. SOM SW1 = OFF-OFF (JTAG). Power on.

```
jtagconfig
quartus_pgm -c 1 -m jtag -o "pvi;top.core.jic"
```

- Power off. SOM SW1 = ON-ON (QSPI boot).

3. Hardware Checklist (every session)

- microSD seated in SOM slot
- SOM SW1 = ON-ON
- Ethernet: Board J6 -> PC Ethernet (direct shared link recommended)
- USB: SOM J2 + Carrier J35 both to PC
- Cameras: Framos FSM:GO IMX678 on MIPI0 and/or MIPI1 (pin 1 to pin 1)
- Display: Carrier DisplayPort TX -> monitor (prefer native DP cable)

IMPORTANT: Do NOT expect /dev/ttyUSB* while the board is powered off. FTDI ports appear only when board is powered.

4. Every Boot - Exact Flow (Direct J6 to PC)

Step A - Boot PC first

- Settings -> Network -> Wired (enp3s0) -> IPv4 -> Shared to other computers

```
ip -4 addr show enp3s0
# need: inet 10.42.0.1/24
```

If not shared:

```
sudo nmcli connection modify "netplan-enp3s0" ipv4.method shared
sudo nmcli connection up "netplan-enp3s0"
ip -4 addr show enp3s0
```

Step B - Power on board

- Power on kit. Wait about 2 minutes for Linux/app start.

Step C - Open HPS serial (new window)

```
ls -l /dev/ttyUSB*
# expect 8 ports when J2+J35 connected

gnome-terminal --title="HPS" -- bash -c "sudo minicom -D /dev/ttyUSB6 -b 115200; exec bash"
```

- Press Enter. Login: root (no password)

NOTE: ttyUSB numbering can change after reboot. HPS is usually the 3rd port of the SOM UART group (often ttyUSB6 when JTAG took 0-3). Nios console on ttyUSB2 showing 'WELCOME TO INTEL NIOS SYSTEM' is normal and is NOT Linux.

If ttyUSB6 is blank, try ttyUSB4 / 5 / 7, or open all and power-cycle once with minicom already open.

Step D - Board IP + MTU 400

```
ip -4 addr show eth0
# expect 10.42.0.x (often .212)

# if no IP:
udhcpc -i eth0
ip -4 addr show eth0

ip link set eth0 mtu 400
```

IMPORTANT: Your PC<->board Ethernet path cannot carry normal 1500-byte frames. Large ping (-s 1472) fails; max working size is about 400. Always set MTU 400 on BOTH sides or the GUI will hang.

Step E - PC MTU + open GUI

```
sudo ip link set enp3s0 mtu 400
sudo ip route replace 10.42.0.0/24 dev enp3s0 src 10.42.0.1 mtu lock 400

BOARD_IP=10.42.0.212   # use eth0 IP from board

ping -c 2 $BOARD_IP
curl -sS --max-time 10 http://$BOARD_IP/ | wc -c
# expect about 1112

google-chrome-stable http://$BOARD_IP/
```

NOTE: Chrome GTK/VAAPI/Wayland messages are normal noise. Success = splash then camera GUI. curl HEAD may return

403; use GET (curl without -I).

Step F - Select camera

- GUI: Input Config -> Input Source -> Camera
- If no live video (especially MIPI1): toggle Camera <-> Input TPG <-> Camera
- Optional log check:

```
grep -iE 'Found camera|Active input|GMSL|No GMSL' /tmp/vvp-isp-demo.log | head -n 30
```

Healthy example: Found camera FSM-IMX678 as Cam 0 / Cam 1; Active input Camera 0; 3840x2160p @60.

5. Serial Port Map (this kit)

- Two Micro-USB cables => usually 8 ttyUSB ports
- One cable group = JTAG/Nios (e.g. ttyUSB0-3). Nios often on ttyUSB2
- Other cable group = HPS UART (e.g. ttyUSB4-7). Linux console = 3rd of that group (often ttyUSB6)
- Settings: 115200 8N1, hardware flow control OFF

```
# open several candidates at once
for i in 4 5 6 7; do
  gnome-terminal --title="ttyUSB$i" -- bash -c "sudo minicom -D /dev/ttyUSB$i -b 115200; exec bash"
done
```

6. Network Notes

Working topology

- Direct: Board J6 <-> PC enp3s0 with Shared networking (10.42.0.1 / 10.42.0.x)

Also tested

- Via switch with static 192.168.100.1 (PC) / .2 (board) - same MTU limit (~400)
- USB-RJ45 dongles - did not fix large-frame problem; some were worse

MTU workaround (required on your lab link)

```
# Board
ip link set eth0 mtu 400

# PC (shared direct)
sudo ip link set enp3s0 mtu 400
sudo ip route replace 10.42.0.0/24 dev enp3s0 src 10.42.0.1 mtu lock 400
```

NOTE: GUI may grey out every few seconds at MTU 400 because WebSocket/UI traffic is fragmented. Live video is on DisplayPort, not the browser - greying is annoying but control usually still works. Fixing large frames (better NIC path) would allow MTU 1500 and a smoother GUI.

Quick MTU health check

```
for s in 56 200 400 500 800 1200 1472; do
  if ping -c 1 -W 1 -M do -s $s $BOARD_IP >/dev/null 2>&1; then echo "OK  $s"; else echo "FAIL $s"; fi
done
```

7. DisplayPort Blank Screen

Cameras can be detected in software while the monitor still shows blank / 'please load image'. That is a DP link issue.

- In GUI first set Input Source = Input TPG. If TPG appears, DP works; switch back to Camera.
- Use carrier DisplayPort TX; prefer native DP cable; avoid weak DP-HDMI adapters for 4Kp60
- Reseat cable; confirm monitor input; try power-cycling monitor
- Nios console (ttyUSB2) may show link-training messages when replugging DP

8. Useful Board Commands

```
ps | grep VvpIspDemo
tail -n 50 /tmp/vvp-isp-demo.log
grep -iE 'Ready|browser|Found camera|Active input|http' /tmp/vvp-isp-demo.log | head -n 40

# local web check on board (BusyBox)
wget -S -O /tmp/local.html -T 5 http://127.0.0.1/
```

```
wc -c /tmp/local.html
# expect 1112 and HTML content

# restart app if needed
killall VvpIspDemo
cd /home/root && ./start.sh
```

9. What NOT to redo every boot

- Do not rewrite the microSD image
- Do not reprogram QSPI (top.core.jic) unless flash was erased or boot is broken

10. Quick Troubleshooting

No ttyUSB devices

- Board powered off, or USB unplugged - power on and reconnect J2/J35

Only 4 ttyUSB ports

- Second USB cable missing - plug both J2 and J35

Login on wrong port / only Nios banner

- Use the other USB group's 3rd port for HPS Linux

Ping works, browser hangs / curl gets headers but 0 body

- Set MTU 400 both sides; confirm with curl | wc -c ~= 1112

IP changed after udhcpc

- Always re-read ip -4 addr show eth0; do not assume .212 forever

Cameras found but no picture

- GUI: Camera <-> TPG toggle; check DP physically

GUI greys out every ~5 seconds

- Likely MTU 400 limitation; live video still on DP; improve Ethernet path later if needed

11. Session Cheat Sheet

```
1) PC Wired = Shared -> 10.42.0.1
2) Power board (SW1 ON-ON)
3) minicom /dev/ttyUSB6 115200 -> root
4) Board: ip -4 addr show eth0; ip link set eth0 mtu 400
5) PC:  enp3s0 mtu 400 + route mtu lock 400
6) Chrome http://BOARD_IP/
7) GUI Input Config -> Camera (TPG toggle if needed)
```

End of guide. Keep this with the kit. Values like ttyUSB6 and 10.42.0.212 are typical for this lab setup but can change after reboot - always verify with ls and ip commands.